GLP-1 abstracts: word embeddings with CBOW and Skip-gram

Project 2, part 2 · Biomedical Information Retrieval

Central question: can about 1,000 scientific abstracts learn a meaningful semantic representation of words?

Short answer: partly. Frequent core terms of the domain form clear, measurable relations in both models: drugs, organs, adverse events. Rare terms are stable only in Skip-gram, and general medical relations that are marginal in the corpus are not learned.

1. Data and preprocessing

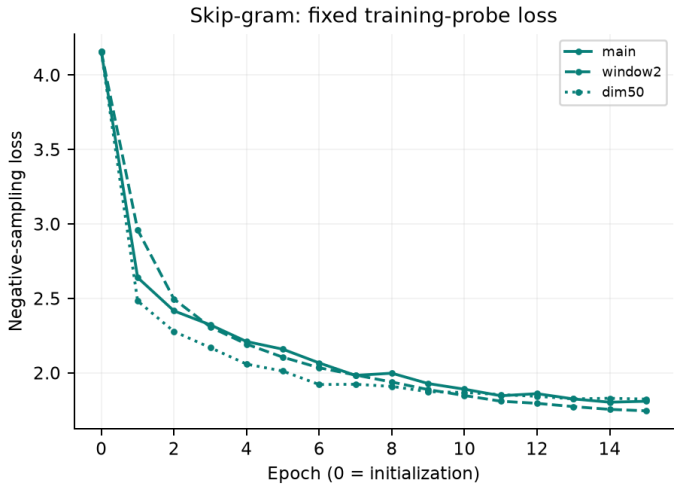
Corpus: the same frozen 1,000 English PubMed abstracts on GLP-1 used in part 1 (corpus fingerprint 43be97743f89d09e…). Only abstract text is used. Preprocessing is part 1's condition C: lowercasing; whitespace tokenization; punctuation stripped from token boundaries while internal hyphens are kept (glp-1); NLTK English stopwords removed; no stemming. Abstracts are split into sentences and context windows never cross a sentence. Result: 10,680 sentences, 177,052 tokens. With min_count 2 the vocabulary has 9,427 words (167,364 retained tokens). Stopwords were removed because they would dominate every window. The threshold of 2 keeps rare domain terms, so their behaviour can be studied; min_count 5 is tested in RQ4.

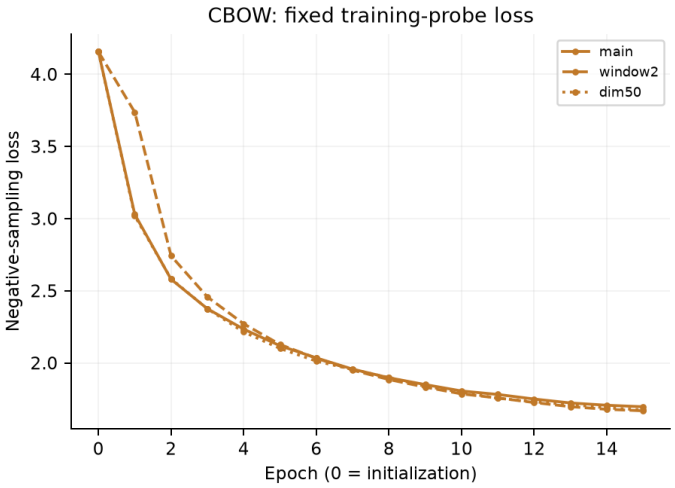
Most frequent words: glp-1 (1606), receptor (1200), patients (1104), weight (1050), obesity (977), metabolic (951), risk (911), agonists (848), diabetes (801), treatment (761), clinical (745), outcomes (742), glucagon-like (741), evidence (739), associated (710), peptide-1 (697), 95% (689), 2 (681), disease (678), effects (653).

2. Training samples and models

Both models use the same window definition (±5 words). Skip-gram creates one (center → context) pair for every word in the window. CBOW creates one (window → center) example per position, with the window's vectors averaged (word2vec cbow_mean=1). Frequent words are randomly subsampled (t = 0.001) each epoch, so example counts vary slightly between runs. Both are trained from scratch in NumPy (src/word2vec.py) with negative sampling: 5 noise words from the unigram^0.75 distribution, learning rate 0.025 decaying linearly, batch 128, 15 epochs, seed 42. No pretrained vectors are used. The loss is measured on a fixed probe of 2,048 training examples and noise words, not on held-out data.

Run	Model	Dim	Win.	Min count	Seed	Vocab	Examples / epoch	Sec.	Probe loss
main	Skip-gram	100	5	2	42	9,427	1,278,168	240	4.157 → 1.810
main	CBOW	100	5	2	42	9,427	159,776	114	4.156 → 1.697
window2	Skip-gram	100	2	2	42	9,427	574,866	138	4.157 → 1.746
window2	CBOW	100	2	2	42	9,427	159,776	85	4.156 → 1.669
dim50	Skip-gram	50	5	2	42	9,427	1,278,153	178	4.155 → 1.824
dim50	CBOW	50	5	2	42	9,427	159,717	89	4.157 → 1.675
mincount5	Skip-gram	100	5	5	42	4,609	1,141,967	218	4.155 → 1.914
mincount5	CBOW	100	5	5	42	4,609	146,027	102	4.153 → 1.674
seed7	Skip-gram	100	5	2	7	9,427	1,278,096	239	4.159 → 1.802
seed7	CBOW	100	5	2	7	9,427	159,780	113	4.158 → 1.748


Skip-gram: fixed training-probe loss
Negative-sampling loss
Epoch (0 = initialization)
main
window2
dim50
4.0
3.5
3.0
2.5
2.0
0
2
4
6
8
10
12
14


CBOW: fixed training-probe loss
Negative-sampling loss
Epoch (0 = initialization)
main
window2
dim50
4.0
3.5
3.0
2.5
2.0
0
2
4
6
8
10
12
14

3. Background: CBOW vs Skip-gram

Property	CBOW	Skip-gram
Input	The context words in the window (their vectors are averaged into one hidden vector h)	The single center word
Prediction target	The center word	Each context word in the window, one at a time
Training examples	One example per token position	One example per (center, context) pair: up to 2 × window per token
Training speed	Faster: fewer, averaged updates	Slower: many more updates per epoch
Rare words	A rare word is one of several averaged inputs, so its own gradient signal is diluted	A rare word is the input for every one of its context pairs, so each occurrence gives it several direct updates
Expected behavior	Smoother, high-cosine space; good for frequent words and interchangeable terms	Sharper, more topical neighbors; better for rare and domain-specific terms, but noisier

Why might Skip-gram be particularly useful for rare or domain-specific scientific terms?

Scientific terms such as drug names, gene symbols or abbreviations usually occur only a few times in a small corpus. In CBOW a rare word only enters training as one of up to 2·window context inputs whose vectors are averaged, so its update is shared with the frequent words around it, and as a prediction target only once per occurrence. In Skip-gram the rare word is itself the input for every pair it forms, so each occurrence produces up to 2·window separate gradient updates that pull it toward the actual words around it. The rare word's vector is therefore determined by its contexts rather than by its random initialization. Its neighbors can be traced to specific contexts, which matters when the term is the thing a biomedical user is searching for.

The experiment agrees (RQ1). Under a seed change, Skip-gram keeps 71% of the top-10 neighbors of words seen 2–4 times; CBOW keeps 30%.

4. Nearest neighbors (cosine, top 5)

15 query words; corpus count in brackets.

Query	Skip-gram	CBOW
glp-1 (1606)	ras 0.63, ra 0.60, polypeptide/glucagon-like 0.57, substitution 0.56, cornerstones 0.55	glp1 0.71, glp-1/gip 0.68, diabetes-labeled 0.63, glp)-1 0.63, croatia 0.60
semaglutide (649)	dulaglutide 0.55, liraglutide 0.48, 2·4 0.44, transitioned 0.43, albiglutide 0.41	tirzepatide 0.92, liraglutide 0.85, dulaglutide 0.82, subcutaneous 0.78, orforglipron 0.77
tirzepatide (359)	dulaglutide 0.54, liraglutide 0.52, exenatide 0.48, favored 0.48, gip/glp-1 0.47	semaglutide 0.92, liraglutide 0.85, dulaglutide 0.84, youth 0.84, subcutaneous 0.81
liraglutide (161)	exenatide 0.58, dulaglutide 0.57, tirzepatide 0.52, injection 0.51, semaglutide 0.48	subcutaneous 0.89, dulaglutide 0.88, orforglipron 0.88, 2·4 0.87, exenatide 0.87
exenatide (23)	lixisenatide 0.85, eudravigilance 0.74, sars 0.73, dose-related 0.69, generalisable 0.68	lixisenatide 0.97, orforglipron 0.94, retatrutide 0.94, cagrilintide 0.93, tzp 0.92
obesity (977)	overweight 0.55, 2030 0.54, pervasive 0.53, 589 0.53, crisis 0.51	increasingly 0.87, osa 0.85, pharmacotherapeutic 0.84, obesity-associated 0.84, apnea 0.84
insulin (397)	resistance 0.74, basal 0.63, beta-cells 0.52, pump 0.50, dd 0.49	basal 0.85, insulin-deficient 0.83, nonobese 0.81, pump 0.79, n-terminal 0.78
nausea (30)	vomiting 0.92, diarrhoea 0.79, hypersensitivity 0.75, hyperhidrosis 0.74, injection-site 0.73	vomiting 0.97, diarrhoea 0.94, infections 0.92, hypersensitivity 0.91, reactions 0.90
cardiovascular (551)	atherosclerotic 0.63, prioritised 0.59, oao 0.58, hospitalizations 0.57, cvd 0.57	cv 0.86, liver-related 0.82, atherosclerotic 0.80, cardiometabolic 0.79, renal 0.79
kidney (309)	ckd 0.63, end-stage 0.59, narrower 0.59, glomerular 0.57, glomerulopathy 0.54	hf 0.92, failure 0.89, heart 0.88, stage 0.86, alzheimer's 0.86
liver (233)	steatotic 0.72, dysfunction-associated 0.65, non-alcoholic 0.62, alcohol-associated 0.61, formerly 0.60	non-alcoholic 0.88, fibrosis 0.88, steatohepatitis 0.87, dysfunction-associated 0.85, ald 0.84
cancer (125)	breast 0.63, endometrial 0.62, cancers 0.60, colorectal 0.58, lynch 0.55	pancreatitis 0.86, cancers 0.85, dependence 0.85, asthma 0.84, nondiabetic 0.82
mice (136)	db/db 0.74, iec-ppara 0.73, diet-induced 0.71, wt 0.71, as21 0.71	diet-induced 0.93, restored 0.90, db/db 0.88, as21 0.88, plasma 0.88
treatment (761)	discontinuation 0.50, assistance 0.47, interindividual 0.44, simplification 0.42, minimum 0.42	adherence 0.86, discontinuation 0.78, patient 0.78, maintenance 0.77, persistence 0.77
diagnosis (72)	10th 0.59, icd 0.59, codes 0.59, classification 0.58, icd-10-cm 0.56	10th 0.90, diagnosed 0.90, pci 0.87, codes 0.86, documented 0.85

Rare words

Query	Skip-gram	CBOW
lixisenatide (8)	exenatide 0.85, eudravigilance 0.79, sars 0.79, generalisable 0.76, phentermine/topiramate 0.75	exenatide 0.97, tzp 0.95, orlistat 0.94, cagrilintide 0.94, low-dose 0.94
diarrhea (7)	constipation 0.88, bloating/distension 0.82, diarrhoea 0.82, deaths 0.81, malabsorption 0.81	constipation 0.94, 4.1% 0.92, 5.4% 0.91, abdominal 0.91, infections 0.91
nafld (11)	non-alcoholic 0.74, 1-point 0.73, called 0.72, 1-stage 0.71, coexists 0.71	dyslipidemia 0.93, non-alcoholic 0.91, presenting 0.91, dehydration 0.91, early-onset 0.90
constipation (8)	diarrhea 0.88, motility-related 0.85, bloating/distension 0.85, deaths 0.83, poisoning 0.81	infections 0.94, diarrhea 0.94, diarrhoea 0.93, 4.1% 0.93, periprosthetic 0.93
rat (10)	organoids 0.82, hk-2 0.80, strain 0.78, agonistic 0.78, transgenic 0.77	nac 0.96, gene 0.95, producers 0.95, competition 0.94, organoids 0.94

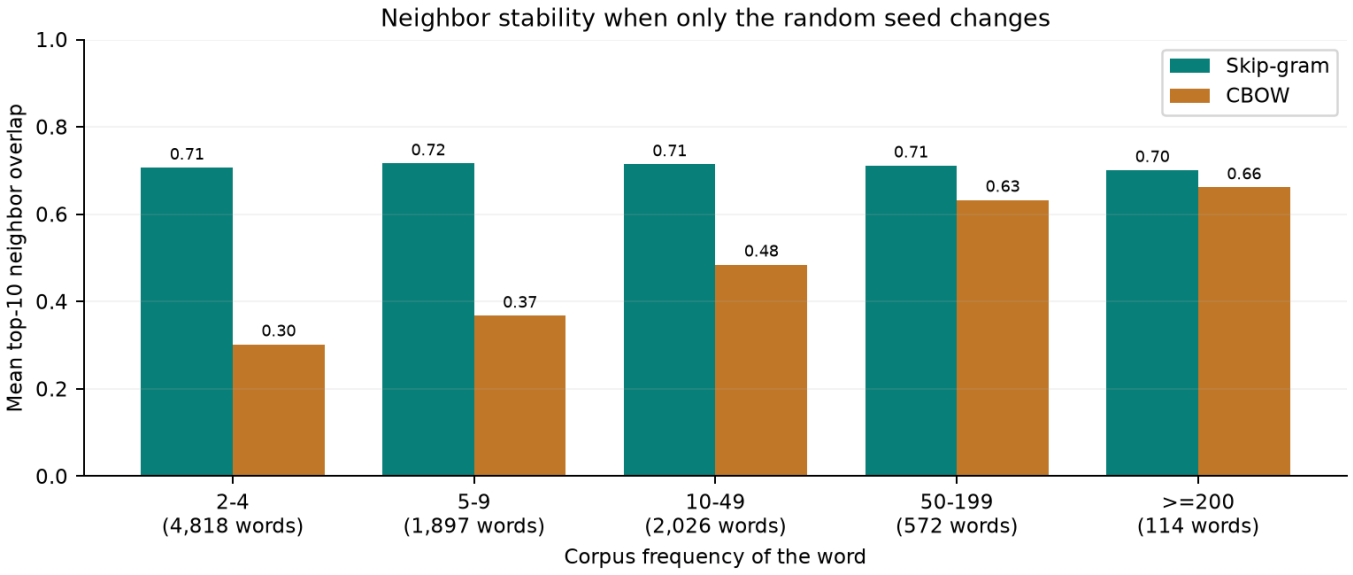
RQ1 — How do CBOW and Skip-gram differ?

Training. With the same corpus and settings (100 dimensions, window 5, 15 epochs), Skip-gram processed 1,278,168 (center, context) pairs per epoch, CBOW 159,776 examples, one per retained token: 8.0× more updates. Skip-gram took 240 s and CBOW 114 s (2.1× slower; all ten models trained concurrently on one machine, so times are comparable to each other only).

Training loss. The fixed training-probe loss fell from 4.157 to 1.810 for Skip-gram and from 4.156 to 1.697 for CBOW. Skip-gram flattens after about ten epochs, while CBOW is still decreasing slowly at epoch 15. The two values are not directly comparable: they score different prediction tasks (one averaged context → word vs. word → each context word). A lower loss is not evidence of a better embedding.

Neighbors. The two models agree surprisingly little. For the 15 query words they share on average 1.5 of 5 nearest words, and across the whole vocabulary only 10% of top-10 neighbors coincide. They also have a different style. CBOW cosines are uniformly high (mean of random word pairs 0.44 vs 0.22 for Skip-gram), and its neighbors are words that fill the same slot in a sentence: semaglutide → tirzepatide 0.92, liraglutide, dulaglutide; kidney → hf, failure, heart (they occur in lists such as 'heart failure and kidney disease'). Skip-gram neighbors are more topical: kidney → ckd, end-stage, glomerular; cancer → breast, endometrial, colorectal; insulin → resistance, basal. But they also include very rare noise words (obesity → '2030', '589').

Rare words. When only the random seed changes, Skip-gram keeps 71% of the top-10 neighbors of words seen 2–4 times. That is about the same as for frequent words (70%). CBOW keeps only 30% for rare words but 66% for frequent words. So CBOW's rare-word vectors are largely an accident of initialization and sampling. Skip-gram's stable rare-word neighbors are not simply memorized from one abstract: only 21% of them ever share an abstract with the word. Both models still connect some rare terms correctly (lixisenatide → exenatide; diarrhea → constipation), but CBOW's rare-word lists also contain numbers such as '4.1%' and '5.4%' with cosines around 0.9.



RQ2 — Are the embeddings semantically meaningful?

Pair	Counts	Skip-gram cosine	pct.	CBOW cosine	pct.
cancer – tumor	125 / 27	0.31	74	0.13	12
treatment – therapy	761 / 486	0.09	20	0.50	56
treatment – diagnosis	761 / 72	0.10	22	0.19	17
cancer – treatment	125 / 761	0.14	31	0.12	12
semaglutide – liraglutide	649 / 161	0.48	95	0.85	97
semaglutide – tirzepatide	649 / 359	0.34	80	0.92	100
obesity – overweight	977 / 98	0.55	98	0.78	92
obesity – weight	977 / 1050	0.35	81	0.51	56
insulin – glucose	397 / 319	0.25	60	0.73	88
glycemic – hba1c	183 / 155	0.26	64	0.71	85
nausea – vomiting	30 / 24	0.92	100	0.97	100
kidney – renal	309 / 171	0.51	96	0.76	91
liver – hepatic	233 / 131	0.43	91	0.71	85
mice – rats	136 / 38	0.55	98	0.83	96
cardiovascular – mace	551 / 73	0.48	94	0.60	70
nausea – kidney	30 / 309	0.10	22	0.45	48
mice – semaglutide	136 / 649	0.22	52	0.30	29
cancer – weight	125 / 1050	0.13	28	-0.04	4

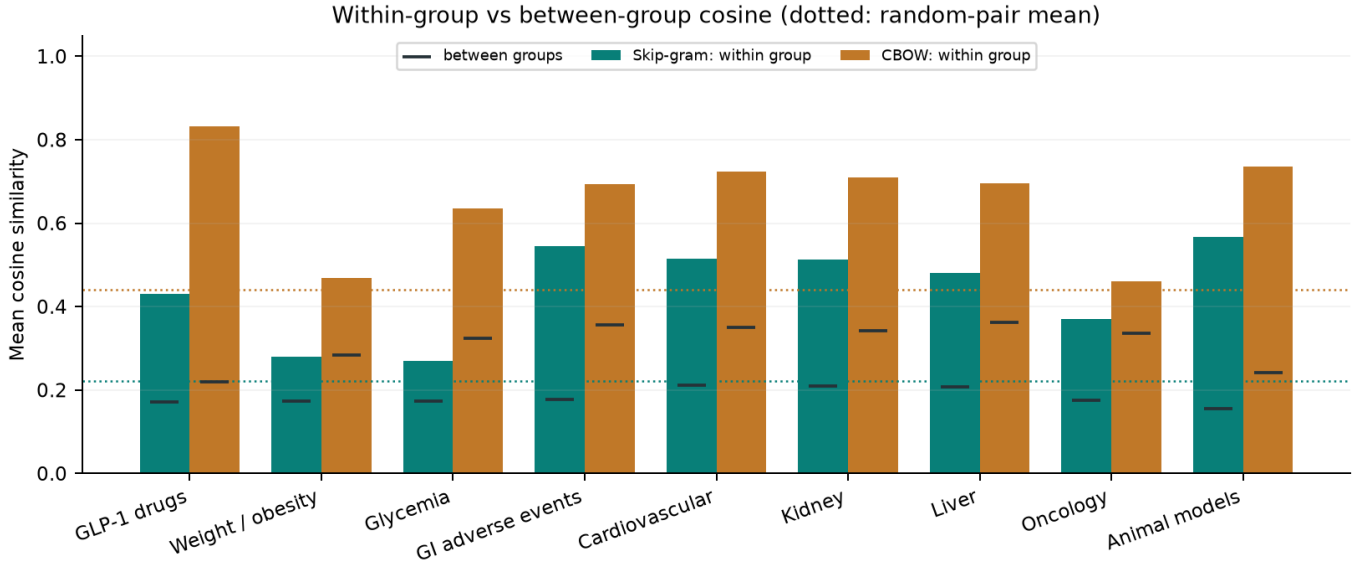
A raw cosine means little on its own, because the two models use the cosine scale differently. Each pair is therefore compared with 20,000 random word pairs (words seen ≥ 5 times) and reported as a percentile. In 5 of 18 pairs, both models put the pair above the 90th percentile: semaglutide – liraglutide, obesity – overweight, nausea – vomiting, kidney – renal, mice – rats. Pairs within nine predefined domain groups are more similar than pairs across groups in every group for both models. The mean margin is 0.26 (Skip-gram) and 0.35 (CBOW) cosine points. Control pairs from unrelated groups stay low: nausea – kidney 0.10 (22th pct.) in Skip-gram, cancer – weight -0.04 (4th pct.) in CBOW.

The brief's example words show the limits. Cancer – tumor scores 0.31 (74th pct.) in Skip-gram but only 0.13 (12th pct.) in CBOW. Treatment – therapy is near-synonymous but scores only 0.09 (20th pct.) in Skip-gram (0.50 (56th pct.) in CBOW). Treatment – diagnosis is low in both (0.10 (22th pct.) / 0.19 (17th pct.)). In GLP-1 abstracts, cancer is a side topic (125 occurrences, 'tumor' 27). 'Treatment' sits next to adherence and discontinuation, and 'diagnosis' next to ICD codes, so these words learn GLP-1-specific contexts rather than general medical meaning.

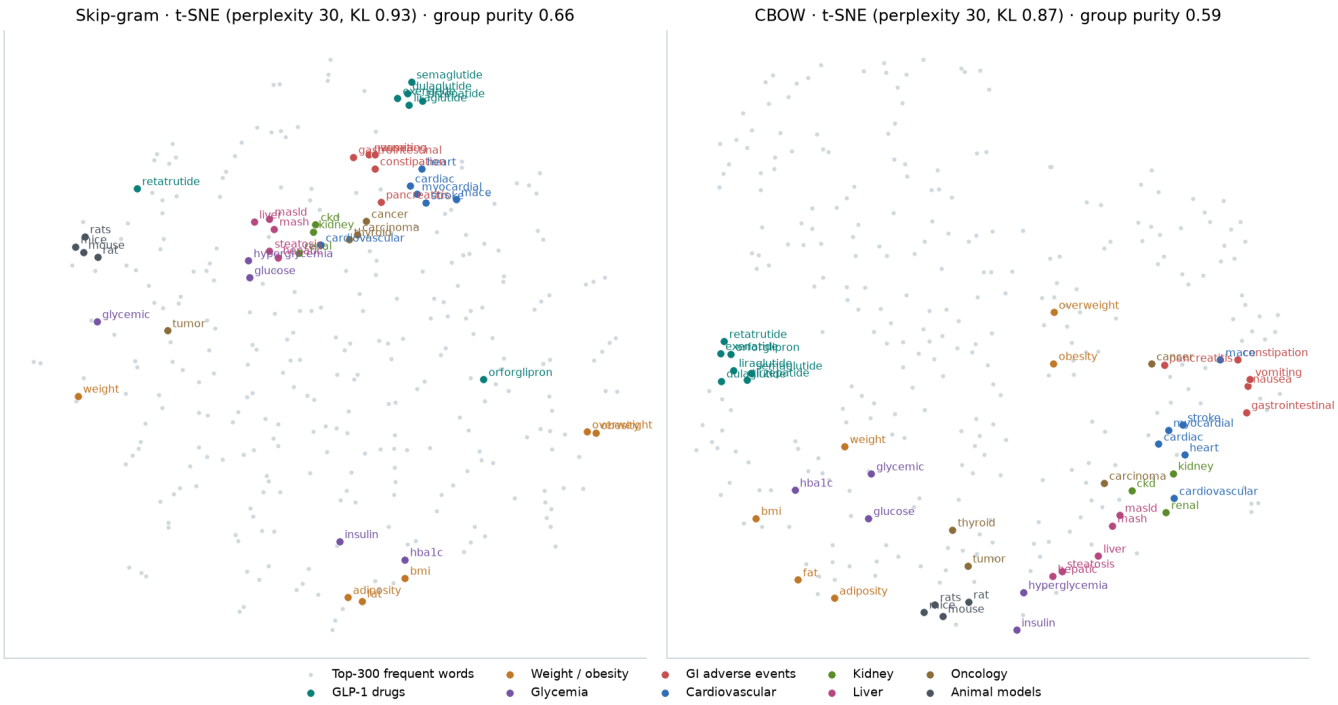
Conclusion: yes, the models learn meaningful relations for well-attested concepts in this domain: drug classes, organ systems, adverse-event terms and animal models. What they learn is relatedness by shared context, not synonymy (kidney ↔ heart in CBOW). The relations are relative to the corpus, and they are unreliable for words that are rare or off-topic in it.

Group	Words	SG within	SG between	CBOW within	CBOW between
GLP-1 drugs	semaglutide, liraglutide, dulaglutide, exenatide, tirzepatide, orforglipron, retatrutide	0.43	0.17	0.83	0.22
Weight / obesity	obesity, overweight, weight, bmi, adiposity, fat	0.28	0.17	0.47	0.28
Glycemia	insulin, glucose, glycemic, hba1c, hyperglycemia	0.27	0.17	0.64	0.32

Group	Words	SG within	SG between	CBOW within	CBOW between
GI adverse events	nausea, vomiting, gastrointestinal, pancreatitis, constipation	0.55	0.18	0.69	0.36
Cardiovascular	cardiovascular, cardiac, heart, mace, stroke, myocardial	0.52	0.21	0.72	0.35
Kidney	kidney, renal, ckd	0.51	0.21	0.71	0.34
Liver	liver, hepatic, masld, mash, steatosis	0.48	0.21	0.70	0.36
Oncology	cancer, tumor, carcinoma, thyroid	0.37	0.17	0.46	0.34
Animal models	mice, rats, mouse, rat	0.57	0.16	0.73	0.24



RQ3 — How do PCA and t-SNE visualizations differ?





PCA is a linear projection onto the two directions of greatest variance. Fitted on the whole vocabulary, it keeps 23.7% (Skip-gram) and 36.1% (CBOW) of the variance of the normalized vectors, so most of the structure is discarded. t-SNE is non-linear. It was fitted on the 300 most frequent words plus all labeled group words, and tries to keep each point's nearest neighbors close.

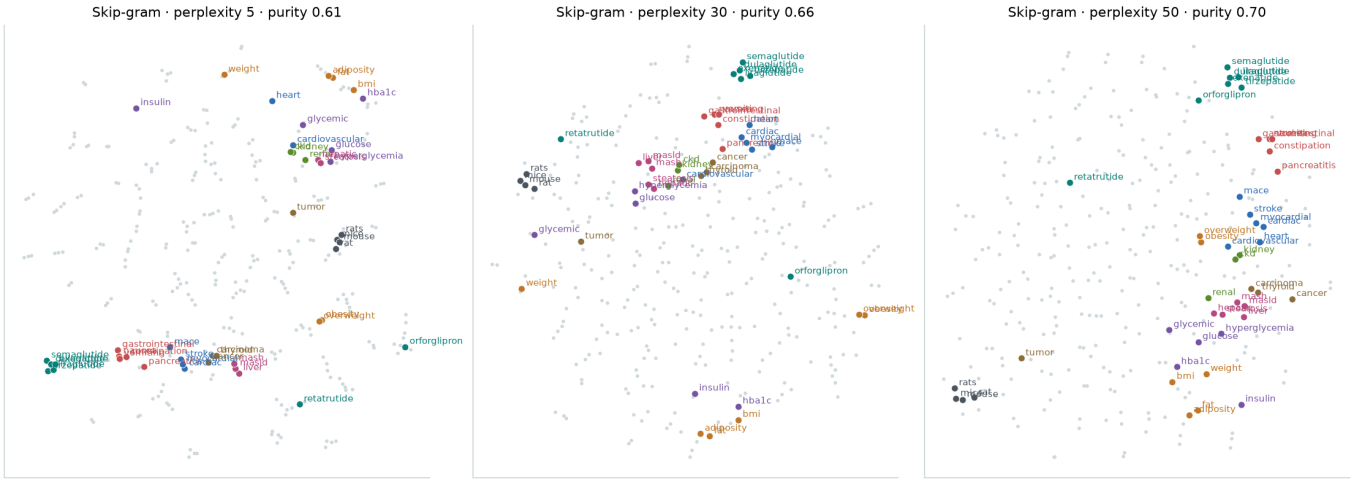
To test clustering numerically, each of the labeled group words is checked against its three nearest labeled words. The table reports the share that belong to the same group (random chance = 0.10). In the original 100-d space this purity is 0.67 (Skip-gram) / 0.59 (CBOW). PCA reduces it to 0.17 / 0.19, barely above chance. t-SNE keeps 0.66 / 0.59, essentially the original value. Visually, the Skip-gram t-SNE shows tight groups for the established GLP-1 drugs, GI adverse events, cardiac terms and animal models. But not every group holds together: renal sits near hepatic rather than kidney/ckd, and the newer agents

orforglipron and retatrutide stay apart from the other drugs. The PCA plots mostly show one dense cloud in which only the animal-model words separate clearly.

Limitations. PCA axes are interpretable and the projection is deterministic and global, but it is too low-rank for 100-d word vectors. t-SNE distances between clusters, and cluster sizes, carry no meaning, and the layout depends on hyperparameters. With perplexity 5, 30 and 50, the same vectors give purity 0.61, 0.66 and 0.70 and visibly different maps. As the brief warns, visual clusters do not prove good semantics. t-SNE can only re-arrange what the vectors already contain, and the clustering is only as good as the cosine purity. A cluster can also form because words share an abstract or a frequency range rather than a meaning. Any claim should therefore be checked in the original space, as done in RQ2.

Model / run	Perplexity	PCA variance	Purity: cosine	Purity: PCA	Purity: t-SNE
skipgram_main	30	23.7%	0.67	0.17	0.66
cbow_main	30	36.1%	0.59	0.19	0.59
skipgram_window2	30	29.8%	0.66	0.18	0.63
cbow_window2	30	35.2%	0.49	0.15	0.41
skipgram_dim50	30	25.3%	0.72	0.13	0.70
cbow_dim50	30	35.3%	0.61	0.21	0.56
skipgram_main_p5	5	23.7%	0.67	0.17	0.61
skipgram_main_p50	50	23.7%	0.67	0.17	0.70

Chance purity: 0.10.



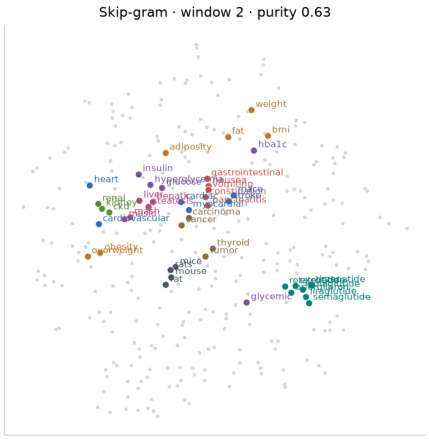
RQ4 — How do model parameters affect the results?

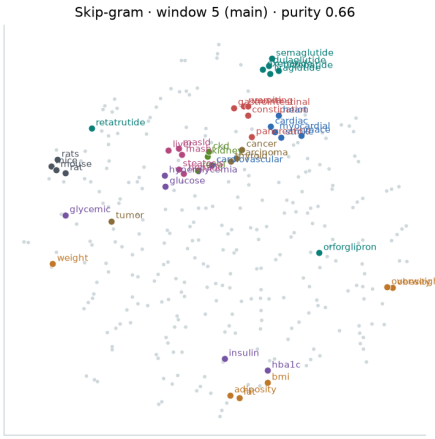
Each variant changes exactly one parameter of the main model. Changing only the random seed sets the noise floor: Skip-gram keeps 71% and CBOW 38% of top-10 neighbors. A parameter matters only if it changes the neighbors clearly more than that.

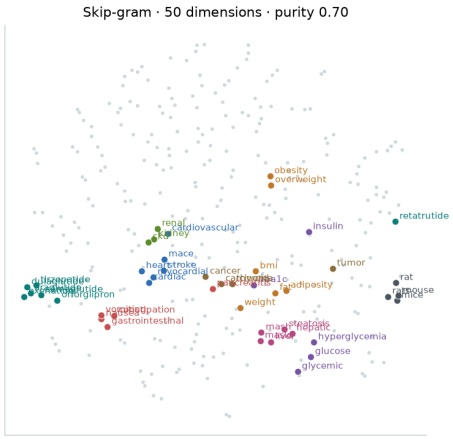
Context window (5 → 2) is by far the strongest factor. Only 19% (Skip-gram) and 8% (CBOW) of neighbors survive. With a narrow window, a word's vector is built from its immediate neighbors (syntactic, same-slot similarity). This raises all cosines (Skip-gram within-group 'GLP-1 drugs' 0.43 → 0.62, but between-group 0.17 → 0.24 too). The mean group margin moves from 0.26 to 0.27 (Skip-gram) and from 0.35 to 0.26 (CBOW). Cosine group purity goes 0.67 → 0.66 (Skip-gram) and 0.59 → 0.49 (CBOW). The window-2 Skip-gram has a lower training loss (1.746 vs 1.810) despite no better groups, which again shows loss is not embedding quality.

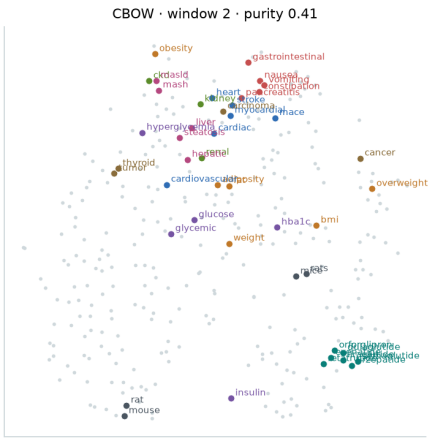
Dimensions (100 → 50) barely matter. Neighbor overlap is 67% / 37%, about the same as the seed-only overlap, and the t-SNE maps look alike. With 167,364 training tokens, 50 dimensions already have enough capacity.

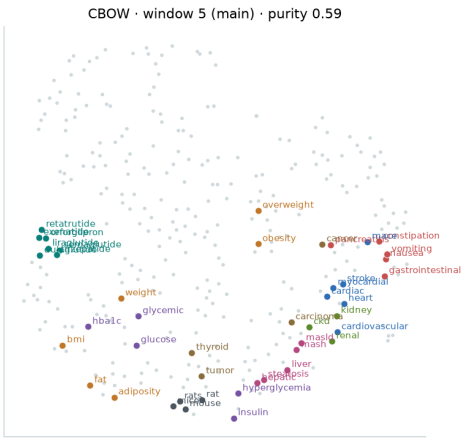
Minimum frequency (2 → 5) halves the vocabulary (9,427 → 4,609) and changes even frequent words' neighbors: only 44% (Skip-gram) / 59% (CBOW) of the top-10 of words seen ≥ 200 times survive. Removing rare words takes noisy candidates out of the neighbor lists (such as '2030' and '589'), and it also removes context words that shaped the remaining vectors.

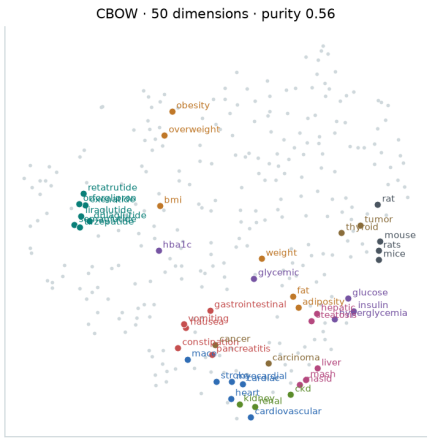












Neighbor overlap by frequency band

Comparison	2–4	5–9	10–49	50–199	≥200	All
Skip-gram | seed 42 vs seed 7 (same setting)	0.71	0.72	0.71	0.71	0.70	0.71
CBOW | seed 42 vs seed 7 (same setting)	0.30	0.37	0.48	0.63	0.66	0.38

Comparison	2–4	5–9	10–49	50–199	≥200	All
Skip-gram | window 5 vs window 2	0.16	0.18	0.21	0.32	0.31	0.19
CBOW | window 5 vs window 2	0.04	0.06	0.12	0.25	0.31	0.08
Skip-gram | 100 vs 50 dimensions	0.67	0.68	0.66	0.64	0.60	0.67
CBOW | 100 vs 50 dimensions	0.30	0.36	0.48	0.63	0.64	0.37
Skip-gram | min_count 2 vs 5	n/a	0.20	0.28	0.37	0.44	0.26
CBOW | min_count 2 vs 5	n/a	0.22	0.36	0.53	0.59	0.33
Skip-gram vs CBOW | main	0.09	0.10	0.12	0.16	0.21	0.10

RQ5 — What are the limitations of a small corpus?

Size. The 1,000 abstracts give 177,052 tokens after stopword removal. Published word2vec models use billions. Of 9,427 vocabulary words (seen ≥ 2 times), 4,818 (51%) occur only 2–4 times, and words seen once are not modelled at all. Rare words are exactly the specialized terms a biomedical searcher may need.

Stability. Even for words seen ≥ 200 times, a different seed changes about 30% (Skip-gram) / 34% (CBOW) of the top-10 neighbors. For rare words, CBOW changes 70%. A neighbor list from one run should be read as a sample, not a fact. Words seen twice regularly enter the top-5 lists with high cosine (glp-1 → 'cornerstones'; obesity → '2030').

Domain. The corpus is about GLP-1, so the space is GLP-1 shaped. 'ras' and 'ra' (receptor agonist abbreviations) are the nearest words to glp-1, 'tzp' means tirzepatide, and general medical relations such as cancer – tumor or treatment – diagnosis are weak because those topics are marginal here. The same words would have different neighbors in an oncology corpus.

Terminology and tokenization. Spelling variants (diarrhea / diarrhoea), punctuation variants (glp-1 / glp1 / glp)-1) and multi-word terms (heart failure, insulin resistance) are split into separate or partial tokens, which spreads the little data each concept has.

Conclusion: 1,000 abstracts are enough to learn coarse, recognisable relations among the frequent core terms of one domain (drug names, organs, adverse events). They are not enough for stable vectors for rare terms, or for general biomedical semantics. Remedies are more text, phrase detection, subword models (fastText), pretrained biomedical vectors (e.g. BioWordVec), and averaging results over several seeds.

Share of top-10 neighbors that co-occur with the word in at least one abstract

Model	2–4	5–9	10–49	50–199	≥200
Skip-gram	0.21	0.30	0.49	0.83	0.96
CBOW	0.10	0.17	0.36	0.76	0.93

Limitations and reproducibility

One topic, one corpus, PubMed order (not a random sample). No external benchmark or retrieval evaluation, so semantic quality is judged from predefined groups, random-pair baselines and seed stability. Groups and pairs were chosen from domain knowledge before the neighbor lists were read. Training times come from concurrent runs. Reproduce with: python scripts/run_project2_part2.py (replays the frozen corpus snapshot; run_metadata.json holds settings and checksums).

Sources

Course brief: project-details-2-part-2-20261006.pdf. Mikolov et al. (2013), arXiv:1301.3781 and arXiv:1310.4546. van der Maaten & Hinton (2008), JMLR 9. Implementation: src/word2vec.py, src/tsne.py, src/embedding_study.py.